Commensurability Classes: The Rational Geometry of Alignment in Event Density Architecture

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Abstract
Collapse behavior in Event Density Architecture (ED-Arch) is governed not only by timing, resonance, recurrence, and scaling, but by the rational structure of the ring's circumference relative to the box length. Law V introduces commensurability classes—equivalence classes of particle diameters d that share the same alignment behavior under drift, recurrence, and resonance. These classes arise from the rational and irrational values of the circumference–box ratio $C(d,N)/L$, which determine whether drift phase returns, nearly returns, or never returns.

Constructive commensurability corresponds to low-denominator rational ratios and produces early collapse, strong ladder adherence, and robust alignment. Destructive commensurability corresponds to high-denominator rational ratios and produces persistent misalignment, repeated window misses, and DECAY. Drift-locked classes arise when the drift phase enters a stable periodic orbit, producing collapse only on specific rungs. Quasi-periodic classes correspond to irrational ratios and produce long-term drift with collapse occurring only through near-miss interactions with recurrence windows.

Law V shows that these four regimes partition diameter space into a commensurability atlas whose structure becomes increasingly intricate as N increases. Under N-rescaling, the atlas converges to universal patterns, revealing that commensurability is the final organizing principle of ED-Arch collapse. Together with Laws I–IV, Law V completes the ED-Arch framework, providing a unified geometric, temporal, resonant, scaling, and classificatory theory of collapse in periodic systems.

1. Introduction
Event Density Architecture (ED-Arch) reveals that collapse in periodic geometries is governed by discrete geometric structures. Laws I–IV established the timing, resonance, recurrence, and scaling behavior of the tangent gate. These laws explain how collapse occurs: through quantized recurrence windows, rational resonance, and the scaling of drift-phase geometry with particle count N.

But one question remains:

> **Which particle diameters collapse, which misalign, and why?**

Empirical sweeps show that collapse behavior is not random. It clusters. Certain values of d consistently produce early collapse, others produce DECAY, others produce long-term drift, and others produce stable periodicity. These clusters are not numerical artifacts — they are the geometric imprint of commensurability.

Commensurability refers to the rational relationship between the ring's circumference and the box length. When these lengths are commensurate, the drift phase returns to alignment; when they are incommensurate, the phase drifts indefinitely. Law V shows that these rational structures partition diameter space into commensurability classes — equivalence classes of d that share the same alignment behavior under drift, recurrence, and resonance.

Law V is the classification law of ED-Arch.

It explains:

- which diameters align constructively
- which diameters misalign destructively
- which diameters produce DECAY
- which diameters produce stable periodicity
- which diameters produce quasi-periodic drift
- how these classes scale with N
- how they organize the collapse landscape

In other words:

Law V provides the taxonomy of ED-Arch collapse.

It is the capstone of the five-law series.

1.1 The Need for a Classification Law

Laws I–IV describe the mechanisms of collapse, but they do not classify the inputs that trigger those mechanisms. Yet empirical sweeps reveal a striking pattern:

- collapse behavior clusters into distinct regimes
- these regimes repeat across N
- the boundaries between regimes follow rational curves
- the structure becomes more intricate as N increases

These patterns demand explanation.

Law V shows that the collapse landscape is organized by the rational structure of:

$$C(d,N) / L.$$

This ratio determines whether drift phase returns, nearly returns, or never returns. The result is a partition of diameter space into commensurability classes that predict collapse behavior.

1.2 Purpose of Law V

Law V provides the first analytic classification of:

- constructive commensurability (early collapse)
- destructive commensurability (DECAY)
- drift-locked classes (stable periodicity)
- quasi-periodic classes (no exact alignment)
- scaling of class density with N
- universal commensurability patterns under rescaling

It explains:

- why certain diameters always collapse early

- why others always produce DECAY
- why some diameters lock into stable cycles
- why quasi-periodic drift persists for irrational ratios
- how these behaviors scale with particle count

Law V is the classification backbone of ED-Arch.

1.3 Relationship to Laws I–IV

- Law I: timing
- Law II: resonance
- Law III: recurrence
- Law IV: scaling
- Law V: classification

Law V unifies the first four laws by showing how timing, resonance, recurrence, and scaling combine to produce distinct commensurability classes. It is the final structural layer of the ED-Arch architecture.

Together, Laws I–V form a complete geometric, temporal, resonant, scaling, and classificatory theory of collapse in periodic systems.

2. Commensurability in Periodic Systems

Commensurability is the geometric principle that governs alignment in periodic domains. Two lengths are commensurate if their ratio is rational. In ED-Arch, the relevant ratio is the circumference of the ring relative to the box length:

$$C(d,N) / L.$$

This ratio determines whether the drift phase returns to alignment, nearly returns, or never returns. Law V begins by formalizing this structure.

2.1 Definition of Commensurability

Two lengths A and B are commensurate if:

$$A / B = p / q$$

for integers p,q.

If no such integers exist, the lengths are incommensurate.

In ED-Arch:

- commensurate ratios produce periodic alignment
- incommensurate ratios produce quasi-periodic drift

This distinction is the foundation of Law V.

2.2 Circumference–Box Ratio

For a ring of N particles of diameter d:

$$C(d,N) = Nd\alpha(N),$$

where $\alpha(N)$ accounts for polygonal spacing.

The key quantity is:

$$C(d,N) / L.$$

This ratio determines:

- drift-phase return
- resonance structure
- alignment behavior
- collapse mechanism

It is the commensurability parameter of ED-Arch.

2.3 Rational vs. Irrational Regimes

If:

$$C(d,N) / L = p / q,$$

then drift phase returns every q cycles.
If the ratio is irrational, drift phase never returns exactly.

Thus:

- rational ratios → periodic alignment
- irrational ratios → quasi-periodic drift

This dichotomy partitions diameter space into commensurability classes.

2.4 Why Commensurability Controls Collapse

Collapse requires:

- recurrence alignment (Law III)
- resonance compatibility (Law II)
- timing compatibility (Law I)

All three depend on whether drift phase returns.

Thus:

Commensurability is the geometric mechanism that determines collapse behavior.

Law V formalizes this mechanism into a complete classification.

3. Constructive Commensurability

Constructive commensurability refers to the set of particle diameters d for which the ring's circumference is in a low-denominator rational relationship with the box length. These values of d produce early collapse, stable alignment, and strong ladder adherence. They are the "good ratios" — the ones that bring the drift phase back into alignment with recurrence windows.

Formally, constructive commensurability occurs when:

$$C(d,N) / L = p / q$$

with small q.
Small denominators correspond to strong periodicity: the drift phase returns frequently, alignment is reinforced, and collapse occurs early.

Constructive commensurability is the geometric mechanism behind the bright bands in collapse surfaces — the regions where collapse is fast, stable, and insensitive to small perturbations.

3.1 Definition

A diameter d belongs to a constructive commensurability class if:

- the circumference–box ratio is rational, and
- the denominator q of the reduced fraction is small.

These classes correspond to strong periodicity:

- drift phase returns every q cycles
- recurrence windows are frequently intersected
- alignment is reinforced rather than disrupted

Constructive classes are the "alignment-friendly" regions of diameter space.

3.2 Collapse Behavior

Diameters in constructive classes exhibit:

- early collapse
- stable rung selection
- minimal DECAY
- robustness to perturbations
- clean, uninterrupted ladders

These are the diameters that "want" to collapse.
They align with recurrence windows and reinforce the timing structure of Law I.

3.3 Window Geometry

Constructive classes correspond to values of d for which:

- drift phase returns near the center of recurrence windows
- alignment is constructive rather than destructive
- the ring enters windows with minimal angular error

Geometrically, these classes sit at the peaks of the recurrence–resonance landscape.

3.4 Scaling with N

As N increases:

- constructive classes become more numerous
- but each class becomes narrower
- their density grows like N
- their spacing shrinks like $1/N$

This matches the scaling laws of Law IV.

Constructive commensurability is not a fixed set — it is a scaling family.

4. Destructive Commensurability

Destructive commensurability refers to the set of diameters d for which the circumference–box ratio is rational with a large denominator. These values of d produce persistent misalignment, window misses, and DECAY. They are the "bad ratios" — the ones that repeatedly push the drift phase out of alignment with recurrence windows.

Formally, destructive commensurability occurs when:

$$C(d,N) / L = p / q$$

with large q.
Large denominators correspond to weak periodicity: the drift phase returns rarely, alignment is fragile, and collapse is repeatedly delayed.

Destructive commensurability is the geometric mechanism behind the DECAY islands of Law II.

4.1 Definition

A diameter d belongs to a destructive commensurability class if:

- the circumference–box ratio is rational, and
- the denominator q is large.

These classes correspond to weak periodicity:

- drift phase returns only after many cycles
- alignment is fragile

- small perturbations cause window misses

Destructive classes are the “misalignment-prone” regions of diameter space.

4.2 Collapse Behavior
Diameters in destructive classes exhibit:

- repeated window misses
- persistent misalignment
- long DECAY intervals
- delayed collapse
- sensitivity to small changes in d

These are the diameters that “resist” collapse.
They sit at the heart of the DECAY structure described in Law II.

4.3 Relationship to Law II
Law II showed that DECAY arises when:

$$\phi(t) \approx 2\pi p / q$$

with $q>1$.

Law V now identifies these q values as destructive commensurability classes.

Thus:

- Law II describes the mechanism
- Law V classifies the inputs that trigger it

Destructive commensurability is the rational backbone of DECAY.

4.4 Scaling with N
As N increases:

- destructive classes become more numerous
- their spacing shrinks like $1/N$
- their density grows like N
- DECAY islands proliferate

This corresponds to the scaling structure of Law IV.

Destructive commensurability is a scaling family that becomes increasingly intricate as N grows.

5. Drift-Locked Classes

Drift-locked classes are the commensurability regimes where the drift phase does not simply align or misalign — it locks into a stable, repeating cycle. These classes sit at the boundary between constructive and destructive commensurability. They are neither strongly periodic nor strongly misaligned. Instead, they produce stable, repeating drift-phase orbits that return to alignment only on specific rungs.

Drift-locked classes are the geometric origin of stable periodicity in ED-Arch collapse. They explain why certain diameters produce collapse only on specific rungs, even when nearby diameters collapse earlier or later.

5.1 Definition
A diameter *d* belongs to a drift-locked class if:

- the circumference–box ratio is rational or nearly rational,
- but the alignment occurs only after a multi-cycle drift-phase orbit,
- and the orbit is stable under small perturbations.

These classes correspond to phase-locked periodicity:

- the drift phase cycles through a repeating pattern
- alignment occurs only at specific points in the cycle
- collapse occurs only on specific rungs

Drift-locked classes are the "periodic islands" of the commensurability atlas.

5.2 Collapse Behavior
Diameters in drift-locked classes exhibit:

- stable periodicity
- collapse only on specific rungs
- repeatable phase-return patterns
- robustness to small perturbations
- neither early collapse nor DECAY

These classes produce the "middle-path" behavior between constructive and destructive regimes.

5.3 Mechanism Geometry
Geometrically, drift-locked classes arise when:

- the drift phase nearly returns to alignment,
- but not exactly,
- and the near-return repeats in a stable cycle.

This produces:

- periodic phase orbits
- stable rung selection
- predictable collapse timing

Drift-locked classes are the fixed points of the drift-phase map.

5.4 Scaling with N
As N increases:

- drift-locked classes become more numerous
- but each class becomes narrower
- their density grows like N
- their stability windows shrink like $1/N$

This matches the scaling structure of Law IV.

Drift-locked classes are a scaling family that becomes increasingly intricate as N grows.

6. Quasi-Periodic Classes

Quasi-periodic classes are the commensurability regimes where the circumference–box ratio is irrational. In these regimes, the drift phase never returns exactly to alignment. Instead, it drifts indefinitely, exploring the phase space without repeating. Collapse occurs only through near-miss interactions with recurrence windows.

Quasi-periodic classes are the geometric origin of long-term drift in ED-Arch collapse.

6.1 Definition
A diameter d belongs to a quasi-periodic class if:

$$C(d,N) / L \notin Q.$$

These classes correspond to irrational drift:

- no exact phase return
- no stable periodicity
- no guaranteed alignment
- collapse only through near-miss events

Quasi-periodic classes are the "irrational seas" of the commensurability atlas.

6.2 Collapse Behavior
Diameters in quasi-periodic classes exhibit:

- no exact alignment
- no stable rung selection
- collapse only through near-miss drift
- high sensitivity to initial phase
- long, irregular collapse times

These classes produce the most unpredictable collapse behavior — not because the system is chaotic, but because the drift phase never repeats.

6.3 Relationship to Recurrence

Law III showed that collapse requires entering a recurrence window.
In quasi-periodic classes:

- the drift phase never returns exactly
- but it comes arbitrarily close
- and eventually intersects a window by near-miss

Thus:

- collapse is guaranteed (for finite windows)
- but the timing is irregular
- and the rung index is unpredictable

Quasi-periodic classes are the irrational limit of recurrence geometry.

6.4 Scaling with N

As N increases:

- rational density increases
- irrational intervals shrink
- quasi-periodic regions become narrower
- but never disappear

This fits the number-theoretic structure of rational approximation.

Quasi-periodic classes are the residual set of the commensurability atlas — always present, but increasingly fragmented as N grows.

7. Discussion

Law V reveals that the collapse landscape of ED-Arch is not a continuous spectrum but a partitioned geometric atlas. Particle diameters fall into distinct commensurability classes, each with its own alignment behavior, collapse timing, and sensitivity profile. These classes are not empirical curiosities — they are the rational and irrational structures that emerge from the circumference–box ratio.

Together with Laws I–IV, Law V completes the ED-Arch architecture.
It shows that collapse behavior is governed by:

- timing (Law I)
- resonance (Law II)
- recurrence (Law III)
- scaling (Law IV)
- commensurability (Law V)

This five-law structure forms a unified geometric theory of collapse in periodic systems.

7.1 Commensurability as the Final Organizing Principle

Commensurability determines:

- whether drift phase returns
- how often it returns
- how strongly alignment is reinforced
- whether recurrence windows are intersected
- whether resonance is constructive or destructive

In this sense, commensurability is the organizing principle that governs the entire collapse landscape. It explains why collapse behavior clusters into distinct regimes and why these regimes repeat across *N*.

Constructive, destructive, drift-locked, and quasi-periodic classes are not arbitrary categories — they are the natural geometric partitions of diameter space.

7.2 Predictive Power

Given *d* and *N*, Law V allows prediction of:

- collapse mechanism (sheet, mirror, tangent, DECAY)
- collapse timing (rung index, delay, periodicity)
- alignment behavior (constructive, destructive, locked, quasi-periodic)
- sensitivity (robust vs. fragile regimes)
- resonance strength (low- vs. high-denominator rationality)

This transforms ED-Arch from a descriptive framework into a predictive system.

7.3 Relationship to the Full ED-Arch Canon

Law V unifies the first four laws:

- Law I explains the timing of collapse.
- Law II explains the resonance structure.
- Law III explains the recurrence geometry.
- Law IV explains the scaling across *N*.
- Law V explains which diameters activate which mechanisms.

This is the final structural layer — the classification that makes the entire system coherent.

7.4 Implications for Periodic Systems

The commensurability classes derived here are not specific to ED-Arch.
They apply to any system where:

- drift interacts with periodic boundaries
- alignment depends on rational relationships
- phase return governs dynamics

This includes:

- coupled oscillators
- lattice-based transport
- periodic scattering systems
- resonant waveguides
- modular arithmetic dynamics

Law V therefore elevates ED-Arch from a domain-specific model to a general geometric framework for periodic systems.

7.5 Law V as the Capstone

With the classification of commensurability classes, the ED-Arch law series is complete.
Law V is the capstone that ties together:

- timing
- resonance
- recurrence
- scaling
- classification

It provides the taxonomy that organizes the entire collapse landscape.

8. Future Work

Law V opens several natural directions for further study. These directions extend the classification framework, integrate it with the scaling laws of Law IV, and prepare the ground for a unified ED-Arch monograph.

8.1 High-Resolution Commensurability Maps

Future work should generate high-resolution maps of commensurability classes for:

- N=3 (baseline)
- N=4,5,6,7,8 (scaling regime)
- N=9–12 (emergent universality regime)

These maps will reveal:

- the density of constructive vs. destructive classes
- the structure of drift-locked islands
- the fragmentation of quasi-periodic regions
- the scaling of class boundaries

Such maps will test the predictions of Law V.

8.2 Analytic Classification of Rational Families

The next step is to derive closed-form expressions for:

- class boundaries

- rational families (p,q)
- scaling of denominators with N
- transitions between commensurability regimes

This will turn the commensurability atlas into a fully analytic object.

8.3 Universal Commensurability Curves
By rescaling diameter space by $1/N$, commensurability classes should collapse onto universal curves. Identifying these curves will unify commensurability across all N.

This is the commensurability analogue of the universal collapse surfaces of Law IV.

8.4 Integration with the ED-Arch Monograph
With Law V complete, the next major project is the ED-Arch monograph, which will include:

- unified notation
- the five laws
- a figure atlas
- universal scaling curves
- commensurability maps
- a glossary of geometric structures

Law V provides the classificatory framework that the monograph will build upon.

8.5 Beyond ED-Arch
Commensurability classes suggest new research directions:

- periodic scattering theory
- rational-phase transport
- modular drift dynamics
- universality in periodic collapse systems

Law V opens the door to a broader mathematical theory of periodic alignment.